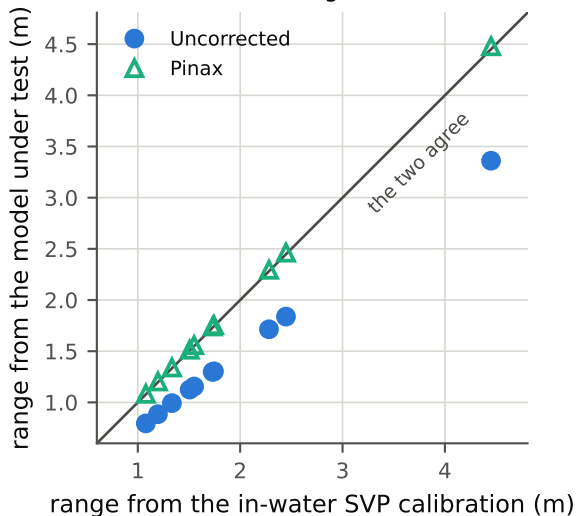


Two independent routes to range, and one that disagrees with both



Ten pool frames, each range taken from the board's own pose. The in-water calibration is fitted to underwater frames and contains no refraction theory; Pinax comes from the in-air calibration and never sees one. They agree to 0.6 %. An uncorrected port sits 25 % below both — the $1/n_w$ the simulation predicts.